ASCE Bridge Designer, Cloud Edition: FAQ for Schools

As of 10 November 2026

This is a compendium of past questions asked by schools and districts. We'll do our best to keep it up to date. In all below, the abbreviation "BD" refers to the ASCE Bridge Designer, Cloud Edition available at asce.org/bridge-designer.

Who are you?

We are a small team of volunteers with no funding and no formal organization. We provide Bridge Designer as-is as a free service to students and teachers. The team is deeply grateful to the American Society of Civil Engineers (ASCE) for hosting the application, however the society does not participate in development or maintenance. See the project at Github for details. Reach us at bridgedesignerteam@gmail.com.

Please provide your VPAT.

The Voluntary Product Accessibility Template (VPAT) is a government standard for communicating the ways a technology supports accessibility for people with disabilities. We have no resources to publish one. ***The Bridge Designer Team seeks volunteers to create and maintain a VPAT for the project and to improve and add accessibility features to the BD itself.*** There are open issues at the Github project. Pull requests are very welcome, but we urge prior coordination.

Does BD require user accounts?

No.

Does BD collect any personally identifying information (PII) about the user?

No.

Does BD collect any information at all?

Only anonymous usage statistics intended to manage ASCE's bandwidth requirements.

Within the user's browser (local and session storage), BD stores only enough information to make the user's design work in progress persistent across browser refreshes. BD doesn't use cookies, though the ASCE's pages may. When so, required warnings and options are provided.

Users store their bridge designs in files on the local machine.

How is BD licensed?

BD is licensed under the GNU General Public License, version 3. Its provisions are spelled out in the file COPYING at the Github project. In a nutshell, it says that anyone can use BD and its source code free of charge for any purpose with one exception. The source code may *not* be copied in any product that isn't published under the same license.

How is the software accessed or installed?

BD is just a web page available by clicking the "Start" button on the landing page at asce.org/bridge-designer or directly at app.asce.org/bridge-designer/app.

How much bandwidth does BD consume?

Tl;dr: not much. The normal compressed size of the BD is less than 1.5 megabytes. This network bandwidth is consumed the first time a user starts the BD. After that, the browser maintains BD in the user's cache. A few things might require a fresh 1.5 Mb download for any given user now and then:

- The user manually clears their browser's cache. Most browsers call this "Delete History."
- The user doesn't touch BD for a long time, in most cases one year.
- The BD team makes a change. Then, a fresh 1.5 Mb copy is provided for the next use.

The upshot is that bandwidth is unlikely to be a problem for modern school networks. An auditorium full of students simultaneously clicking the BD link for the first time might cause a backlog of a minute or two.

What kind of device and software does BD require?

There are only three requirements:

- Screen of at least 1280 by 720 pixels.
- Accurate mouse-like pointer including mouse, touch pad, track ball, or similar.
- Fairly up-to-date version of most modern browsers (Safari is an exception).

Notes:

- A large-enough pad will work with a stylus that looks like a pointer to software.
- Touch-only devices aren't supported.
- Most phones don't have enough pixels.
- If your device doesn't meet the requirements, you'll see a warning that you can override to try using the BD anyway.

What is known about the user community?

The BD implementation as a web app is new. In the first four months of use, we saw about 120,000 uses by 24,000 people. BD is based on a very similar desktop app that was available 2001 to 2026 in various versions, downloaded at least 4 million times, but probably many more via anonymous mirrors. We know of users in 40+ countries, many colleges, school districts, schools, and home school groups. Students have included pre-K through PhD. Most questions and comments, which we welcome, come from engineers and teachers of engineering, physics, technology and science - along with their student users.

What is known about student usage patterns (“screen time”)?

In classroom settings, K-12 students who are already skilled computer users produce a successful design - supporting the truck load - in less than a half-hour. Younger students generally require more coaching, including use of the BD’s template feature.

Searching for an optimal design can be time-constrained or open-ended. For an optimal version of an initial design, another half-hour suffices. Consequently, it’s feasible for a teacher to conduct a contest to find the best bridge within a single one-hour class period, especially with more advanced students. If a teacher seeks to limit time spent by students on optimization, they specify a single site condition via the local contest code feature and an appropriate deadline - hours or days - for submitting results.

On the other hand, finding a near-optimum design after considering every reasonable truss configuration for every site condition involves several hours of focused work that includes gaining deep insight into the ways trusses carry loads. Many teachers conduct extended contests over days or weeks, allowing interested students to maximize their learning about truss structures. The BD’s help on how to design a bridge supports this.

Bug reports and feedback

The team responds to bridgedesignerteam@gmail.com but prefers issue reports submitted at the project Github. ***We appreciate feedback of all kinds.***

How about data privacy and security?

Since BD collects no data and stores information only to the user’s machine locally, it presents no data privacy threat. ASCE’s web hosting follows industry best practices. Since they store no user data, their only significant risk regarding the BD is “denial of service.” I.e., the BD would be unavailable if they experience down time, which isn’t a privacy issue. ASCE uses Google Analytics to collect aggregated, anonymous usage data.

Does the BD use artificial intelligence?

No. BD is designed to inspire students by providing a series of independent discovery opportunities - “aha moments” - as they master increasingly sophisticated aspects of engineering design. At present, we believe AI can only dilute that experience. ***If you disagree, please add a comment to the master issue for AI features.*** We welcome your ideas! If we ever do incorporate AI, future versions of this document will explain.

Will you certify BD complies with specific requirements?

No. The app is provided as-is. We do certify that this document is true and correct to the best of our knowledge as of its publication date.